

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary Summer Examination – 2025**

**Course:** B. Tech      **Branch:** Civil Engineering      **Semester :** VI

**Subject Code & Name:** BTCVC601\_Y18 & Design of Concrete Structures I

**Max Marks:** 60

**Date:** 30/06/2025

**Duration:** 3 Hr.

**Instructions to the Students:**

1. Each question carries 12 marks.
2. **Question No. 1 will be compulsory and include objective-type questions.**
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

|             |                                                                                                                                   | (Level/C<br>O) | Mark<br>s |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| <b>Q. 1</b> | <b>Objective type questions. (Compulsory Question)</b>                                                                            |                | <b>12</b> |
| 1           | Minimum grade of concrete for PCC is -----<br>a.M-20      b. M-25      c. M-15      d. M-10                                       | 1              | 1         |
| 2           | Minimum no. of bars to be used in circular column is -----<br>a. 2      b.3      c. 4      d.6                                    | 1              | 1         |
| 3           | Minimum diameter of bar used in steel reinforcement is-----mm.<br>a. 3      b. 5      c. 6      d. 8                              | 2              | 1         |
| 4           | Minimum diameter of bar to be used for column is -----mm<br>a.10      b. 12      c.16      d.20                                   | 2              | 1         |
| 5           | ( $x_u/d$ ) limiting value for Fe-415 is -----<br>a.0.53      b.0.48      c. 0.55      d. 0.46                                    | 3              | 1         |
| 6           | Beam is design as doubly if $M_u$ ----- $M_{u,limiting}$<br>a. equal to      b. more than      c. less than      d. none of these | 3              | 1         |
| 7           | Single flight of a dog-legged stair is -----slab<br>a. one way      b. two way      c. cantilever      d. none of these           | 4              | 1         |
| 8           | $M_{u, limiting}$ for Fe-415 is ----- $f_{ck}bd^2$<br>a.0.133      b. 0.138      c. 0.148      d. _____                           | 4              | 1         |
| 9           | Vertical stirrups in beams resist -----<br>a. B.M      b. SF      c. Torsion      d. All of these                                 | 5              | 1         |
| 10          | LL for public building is -----kN/m <sup>2</sup>                                                                                  | 5              | 1         |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                           |                    |                        |   |    |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------|------------------------|---|----|
|                                      | a.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | b.2                       | c. 3               | d. 46                  |   |    |
| 11                                   | Position of main steel in a cantilever beam to resist BM is at ----                                                                                                                                                                                                                                                                                                                                                                                                  |                           |                    |                        | 6 | 1  |
|                                      | a. bottom mid section                                                                                                                                                                                                                                                                                                                                                                                                                                                | b. bottom support section | c. top mid section | d. top support section |   |    |
| 12                                   | Load carrying capacity of a rectangular column is -----% less than that of circular column.                                                                                                                                                                                                                                                                                                                                                                          |                           |                    |                        | 4 | 1  |
|                                      | a. 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                | b.15                      | c.5                | d. 25                  |   |    |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                           |                    |                        |   |    |
| Q. 2 Solve the following.            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                           |                    |                        |   | 12 |
| A)                                   | What do you mean by over- reinforced beam?<br>Design by LSM, a simply supported beam, 3 m in span subjected to u.d.l of 20 kN/m over its entire span and a central point load of 15 kN. Use M-25 grade concrete, Fe-415 grade steel and 20 mm diameter bars. Adopt width of beam equal to 300 mm. Given $d'=25$ mm<br>.                                                                                                                                              |                           |                    |                        | 1 | 6  |
| B)                                   | Explain role of longitudinal steel and lateral ties in columns.<br>Design a short RCC column to carry an axial ultimate load of 1000 kN. Clear height of column is 3.1 m. Use M-20 grade concrete and Fe-415 grade steel. Draw a neat diagram of column (longitudinal section and cross-section) to show reinforcement details.                                                                                                                                      |                           |                    |                        | 4 | 6  |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                           |                    |                        |   |    |
| Q.3 Solve the following.             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                           |                    |                        |   | 12 |
| A)                                   | Design by LSM a slab for a passage 3.1 m wide, supported over two opposite beams, each 230 mm wide. The LL & FF on the slab is 3kN/m <sup>2</sup> and 0.5 kN/m <sup>2</sup> respectively. Use M-20 grade concrete and Fe-415 grade steel. Draw neat sketches to show reinforcement details in cross-sections (short span and long span) and top view.                                                                                                                |                           |                    |                        | 3 | 6  |
| B)                                   | Design a dog-legged stair for floor to floor height 3.2 m. Take LL and FF equal to 3 kN/m <sup>2</sup> and 1 kN/m <sup>2</sup> . Use M-20 grade concrete, Fe-415 grade steel and 20 mm diameter bars. Width of each flight is 1.0 m. Draw neat diagrams to show reinforcement details in longitudinal section and plan.                                                                                                                                              |                           |                    |                        | 5 | 6  |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                           |                    |                        |   |    |
| Q. 4 Solve Any Two of the following. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                           |                    |                        |   | 12 |
| A)                                   | Under what situations a doubly reinforced beam is adopted?<br>Design by LSM ,a simply supported beam, 5.5 m in span subjected to u.d.l of 60 kN/m over its entire span. Use M-25 grade concrete, Fe-500 grade steel and 16 mm diameter bars. Adopt Size of beam equal to 300 mm x 500 mm (Overall). Given $d'=30$ mm for compression and tension reinforcements. Draw a neat diagram of beam (longitudinal section and cross-section) to show reinforcement details. |                           |                    |                        |   | 6  |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |           |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|
| <b>B)</b>   | Explain structural behavior of a one-way slab.<br>Design by LSM, a slab for a room 3m x 4m (inner dimensions) supported over four beams, each 300 mm wide. The LL & FF on the slab is 2 kN/m <sup>2</sup> and 0.5 kN/m <sup>2</sup> respectively. Use M-20 grade concrete and Fe-415 grade steel. Assume slab is continuous over all edges. Draw neat sketches to show reinforcement details in cross-sections (short span and long span) and top view.                             | <b>4</b> | <b>6</b>  |
| <b>C)</b>   | A simply supported beam, 3 m in span subjected to u.d.l of 20 kN/m over its entire span. Use M-20 grade concrete, Fe-415 grade steel and 12 mm diameter bars. Adopt size of beam equal to 300 mm x 400 mm (overall). Given d'=25 mm. Beam is reinforced with 3 bars of 16 mm at support and mid -section. Design the beam for shear reinforcement. Draw neat diagrams to show reinforcement details in longitudinal section and cross-section. Adopt <b>working stress method</b> . | <b>3</b> | <b>6</b>  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |           |
| <b>Q.5</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              |          | <b>12</b> |
| <b>A)</b>   | Design by LSM, a square footing for a column 300mm x 300mm to carry an axial ultimate load of 400 kN. SBC of soil is 400 kN/m <sup>2</sup> . Use M-20 grade concrete and Fe-500 grade steel. Draw a neat diagram to show reinforcement details (plan and longitudinal sections)                                                                                                                                                                                                     | <b>6</b> | <b>12</b> |
| <b>B)</b>   | A simply supported beam, 3 m in span subjected to u.d.l of 20 kN/m over its entire span. Use M-20 grade concrete, Fe-415 grade steel and 20 mm diameter bars. Adopt size of beam equal to 300 mm x 400 mm (overall). Given d'=25 mm. Beam is reinforced with 3 bars of 16 mm at support and mid -section. Design the beam for shear reinforcement. Draw neat diagrams to show reinforcement details in longitudinal section and cross-section. Adopt <b>limit state method</b> .    | <b>4</b> | <b>6</b>  |
| <b>C)</b>   | Explain the procedure for design of shear for a beam .Draw neat sketch to support your answer.                                                                                                                                                                                                                                                                                                                                                                                      | <b>5</b> | <b>6</b>  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |           |
| <b>Q. 6</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>5</b> | <b>12</b> |
| <b>A)</b>   | Explain with neat sketches the term: Development length                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>4</b> | <b>6</b>  |
| <b>B)</b>   | Explain the Structural behavior of two-way slab with sketches                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>3</b> | <b>6</b>  |
| <b>C)</b>   | Write step by step procedure for design of a footing.                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>3</b> | <b>6</b>  |
|             | <b>*** End ***</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |           |